

A dark-themed banner for CS265. It features various icons related to data science and systems: a cloud with an infinity symbol, a gear, a laptop, a smartphone, a magnifying glass, an envelope, a server rack, a padlock, and a line graph. The text 'CS265' is prominently displayed in the center, with 'CS' in red and '265' in white. Below it, the text 'Data Systems Research for the Big Data era' is written in white.

CS265

Data Systems Research for the Big Data era

[HTTP://DASLAB.SEAS.HARVARD.EDU/CLASSES/CS265/](http://daslab.seas.harvard.edu/classes/cs265/)

CS 265 SYLLABUS

Prof: Stratos Idreos (stratos@seas.harvard.edu)

After taking CS265, you will be able to understand big data and AI system internals for LLMs, Machine Learning, and broad Data Science tasks, as well as have a handle on what it means to do CS systems research.

WHAT IS THIS CLASS ABOUT?

Big data is everywhere. A fundamental goal across numerous modern businesses and sciences is to be able to utilize as many machines as possible, to consume as much information as possible and as fast as possible. The big challenge is "how to turn data into useful knowledge and actionable decisions". This is a moving target as both the underlying hardware and our ability to collect data evolve. In this class, **we will discuss how to design systems (data structures and algorithms)** for key data-driven areas, including **NoSQL, newSQL, LLMs, machine learning, RAG, Image AI, and neural networks**. We will see how they all rely on the same set of very basic concepts and we will learn how to synthesize efficient solutions for any problem across these areas using those basic concepts.



WHAT IS A DATA SYSTEM & WHAT IS AN AI SYSTEM?

Data systems and AI systems are literally everywhere. We are using them directly or indirectly every day all day long for numerous basic or not so basic tasks, e.g., when we are buying coffee to when we are booking airplane tickets or training a neural network. They provide the backbone of all modern businesses to manage their data and decisions and of course they provide the backbone of online businesses and environments such as social networks and search engines. They are also used increasingly in science as data becomes more and more the fundamental enabler in generating knowledge.

WHAT IS THIS CLASS NOT ABOUT?



This class is not a traditional introduction on how we use a database systems, AI systems, and how to write SQL or Python code. Instead, this is a systems class about system design. You will learn how big data and AI systems work at their core and how to design new systems for emerging data-driven applications and hardware. By the way, if you know how systems work, you also become better at using them!

WHY TAKE THIS CLASS?

Data is everywhere. Every year we create even more data. As it stands, every two days we create as much data as much we created from the dawn of humanity up to 2003 [Eric Schmidt, Google]. Sciences, businesses, and everyday life are substantially affected. Data systems are in the middle of all this. Data systems are how we store and access data, i.e., they are the backbone of any data-driven application. It is a \$100B industry, growing 10% every year [Economist, “Data, data everywhere”].

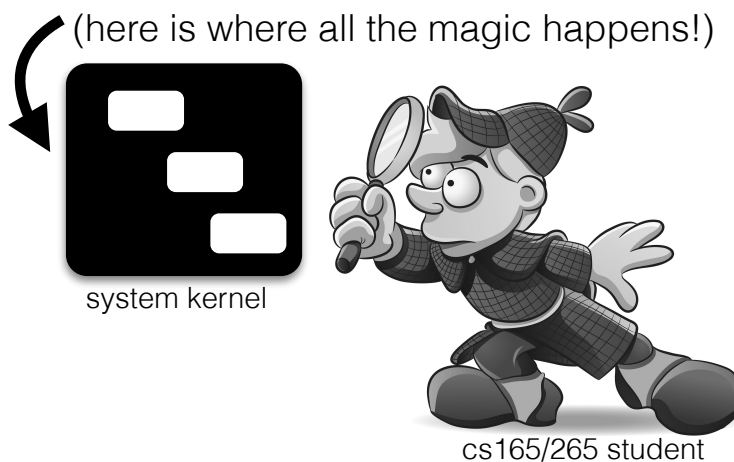
At the same time, systems research and the whole industry are undergoing a major, ongoing transition; as new data-driven scenarios and applications emerge, there is a constant need to redefine what constitutes good system design in such a dynamic environment.

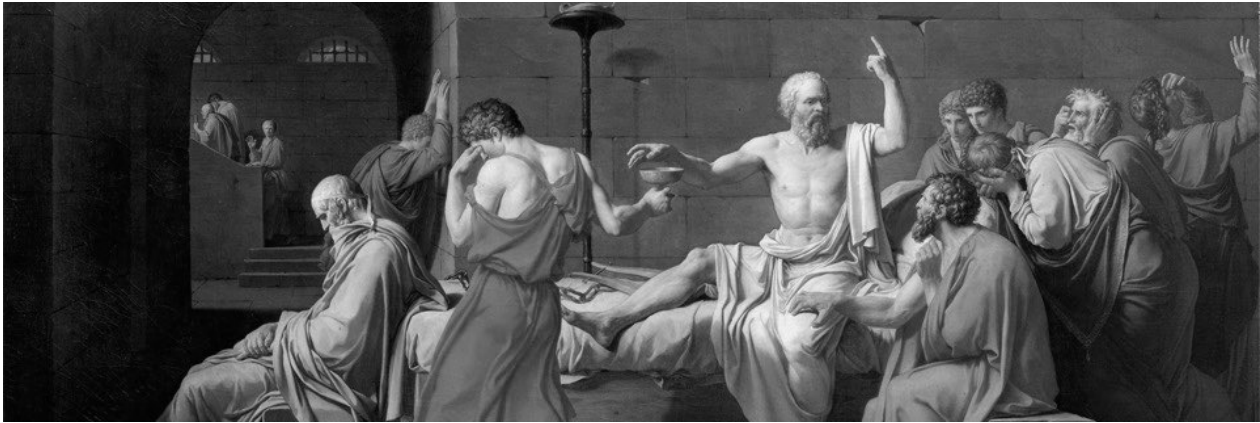


CS265 exposes students to the core internals of data & AI systems, enabling them to understand key trends in system design and become among the few who can design and evaluate systems. In addition, due to the way the course is taught (focus on interactive problem solving, open topics, and the latest research results), this is also a great class for those who want to understand what CS research is all about and how to engage in doing research.

WHAT IS THE EXPECTED LEARNING OUTCOME?

1. Learn state-of-the-art research and industry trends in big data & AI systems.
2. Understand the tradeoffs in designing and implementing modern systems.
3. Be able to make design decisions in big data-driven scenarios.
4. Understand the fundamental principles that govern all systems and how these apply across diverse areas: NoSQL, LLMs, Neural Networks, Data Science, Vision.
5. Develop basic research skills: reading, writing, and understanding research papers.
6. Deepen C programming, debugging, and performance profiling skills.





CLASS PHILOSOPHY

CS265 has unlimited office hours, relies on the latest research papers instead of a standard textbook, lectures are based on interaction and discussion instead of just “lecturing”, many of the quizzes and problem sets are actually open research problems, and most of all, it is fun! The instructor and the TFs are here to help you every day and at all times throughout the semester. You may request as many meetings as you like and as much help as you want.

The course is also geared toward engaging students in creative thinking and problem-solving to give them a sense of how computer science research takes place. Many of our students have successfully conducted research projects with [DASlab](#) and published papers.

HOW DOES 265 COMPARE TO 165?

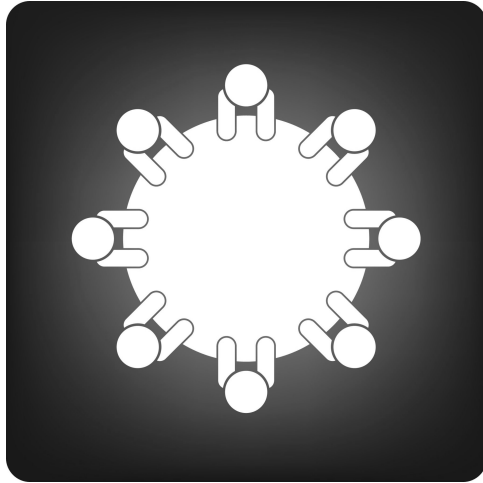
If you took and liked CS165, you will like CS265 as well. From a content standpoint, CS265 builds on CS165 by covering additional topics, with a focus on AI. In terms of how the class is taught, it is even more interactive and research-oriented. Semester projects focus on open research problems with the potential to lead to publication, and every class is focused on a single research paper and its detailed understanding.

HOW MUCH WORK IS IT?

You may have heard stories about CS165 and are wondering whether CS265 will be equally hard, or you may have taken CS165 and are wondering whether this will be a similar amount of work. CS165 and CS265 are different styles of classes. While CS165 is much more focused on implementation, leading to a full system prototype, CS265 is more focused on ideas and design. In other words, you may have written 5-10K lines of code (some even more!) for CS165 but in CS265 you are more likely going to write small amounts of code and mostly play with alternative ways to design a specific functionality, structure or algorithm or model to highlight the effect of different choices and to find out new ways to solve a specific problem.

LECTURES

The class meets twice a week.



INTERACTION IN EVERY CLASS

While the instructor will give a few lectures throughout the semester, the class will be primarily discussion-based. Think of this as an extended brainstorming session, a round table discussion about a specific problem in each class. The goal is to maximize interaction.

Our discussion will aim to highlight design trends and trade-offs, as well as algorithmic issues. Another significant part of our discussions will focus on examining open problems and highlighting opportunities for innovation.

At the very beginning of the semester, the instructor will do 4-5 lectures to provide the necessary background. After that, each class will be based on a student presentation about a recent research paper, which will work as a trigger for the day's brainstorming. Depending on the class's needs, the instructor will give additional lectures during class time or during our extra research sessions.

OFFICE HOURS & LABS

Interaction does not stop at lecture time. CS265 is designed to maximize interaction, as we truly believe this is the best way to learn; we offer daily office hours and labs.

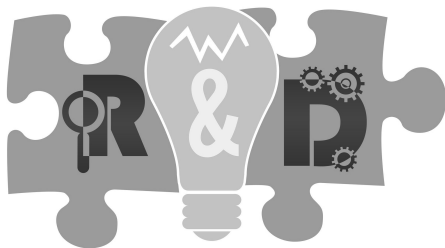
Starting Week 1, Prof. Idreos will hold office hours during the week, and additional OH will be offered periodically during the weekend. Labs are offered by the TFs. Rooms and slots: TBA on the class website. The goal of labs is to get hands-on help for the projects (coding). Bring your laptop and any questions about specific parts of the project you need help with. Labs are the place to go when you have a persistent bug, when you need help with a specific tool for the project (e.g., for debugging or performance testing), or to get feedback about the quality of your coding.

CLASS RECORDINGS

All classes and interactive sessions will be recorded and available online. Even if you miss a class, it will be easy to catch up, and you can use these recordings to review specific material throughout the semester (e.g., to prepare for midterms).

SECTIONS

Another component of the course is sections. Sections are used to deliver material about the class, i.e., to go more deeply into some of the concepts discussed in class, to do additional quizzes, or to deliver background material that is needed to follow next week's class or for the project. There will be no actual section meeting. Instead, all sections will be recorded by the teaching staff, and videos will be posted online. The material posted will be tailored to provide a step-by-step guide for each topic, making it easy to follow without being physically present in the section. However, if there are still questions about the material presented in the sections, you can ask them during the daily office hours or in labs.



RESEARCH SESSIONS

Throughout the semester, on select days, the instructor and DASlab PhDs and postdocs will discuss research! First, DASlab researchers will present their recent work in data systems and connect it to the material you are learning in class. Then you will have the chance to talk with them about their research and open problems, and to learn about open research opportunities. Snacks and drinks will be provided.

WEEKLY REVIEWS

Each student will write two paper reviews per week. This prepares you for the class discussion. Reviews should be no more than two pages long.

Each review should have text for at least the following 9 points:

- 1) what is the problem?
- 2) why is it important?
- 3) why is it hard?
- 4) why existing solutions do not work?
- 5) what is the core intuition for the solution?
- 6) does the paper prove its claims?
- 7) what is the setup of analysis/experiments? is it sufficient?
- 8) are there any gaps in the logic/proof?
- 9) describe at least one possible next step.

Reviews are not required and are not graded. Their utility lies in preparing students for the class discussion.



PRESENTATIONS

Each student will do at least one paper presentation during the semester. Presentations should follow similar guidelines as the guidelines for reviews. There should be 1-2 slides for each one of the nine core points in the review guidelines. In addition, there should be detailed slides that describe the core idea of the paper.

Your slides should not be a multiple sheets of bullet lists - in fact try to avoid bullet lists altogether - your slides should follow the generic formatting you will see in the first four lectures, that is: make slides as simple as possible - avoid text unless absolutely needed - no full phrases unless you need to give an exact definition of something - use figures and visual examples, one slide one message=each slide should have a single goal that you should be able to describe within a single phrase.

Your slides should be reviewed by the instructor at least 24 hours before the class you are presenting. The final deck of slides should be available 30 minutes before class so we can upload online. You are welcome to join for OH for help >>1 while you prepare your slides!



WHO CAN TAKE THIS CLASS?

IF you have taken CS165, CS161, CS261
GOTO next syllabus section;
ELSE
see below;



Background: Naturally, the more background you have, the smoother your experience in 265 will be. Prior knowledge of C programming and systems programming, as well as a solid understanding of computer architecture, particularly the memory hierarchy (including cache memory), is essential for this class. Courses that provide a systems background (such as CS50 and, in particular, CS61 or equivalent) are essential. Good hacking, algorithm designing, and data structures skills are also required.

If you are a graduate student and have taken a mix of systems (database, operating, and distributed systems) classes in the past, you will be OK, and we will provide enough background for you to follow. CS265 satisfies the systems requirement for a PhD.

If you are a senior in college and this is your last chance to take this class, if you have taken CS61 but no CS161 or CS165, then talk to the instructor to evaluate how fit you are for the class. If you have not taken CS61 but do have significant systems programming experience, you may still qualify.

In all other cases, you can take the class as the “systems track” instead of the “research track”. This simply means that you start with the systems project instead of the research project until you build enough systems background.

HOW CAN I DO GREAT IN 265?

Just utilize all resources provided. Show up in class to participate in interactive sessions. There are also daily office hours and labs; show up as often as possible so we can help with anything you need! When you find yourself stuck with the project, either with a design decision or just a bug, it is normal to struggle for a while — it is part of the learning process — but after some time, grab your laptop and come by!



WHAT CAN I DO TO PREPEARE?

Especially if you have not taken CS165, it is a good idea to spend some time preparing before the semester starts and during the early weeks of the semester, even if you consider yourself an expert systems student. The best approach is to browse some foundational readings on data systems architecture. We propose that you take a look at the following texts from the CS165 readings:

1) Get familiar with the very basics of traditional database architectures:

Architecture of a Database System. By J. Hellerstein, M. Stonebraker and J. Hamilton. Foundations and Trends in Databases, 2007

2) Get familiar with very basics of modern database architectures:

The Design and Implementation of Modern Column-store Database Systems. By D. Abadi, P. Boncz, S. Harizopoulos, S. Idreos, S. Madden. Foundations and Trends in Databases, 2013

3) Get familiar with the very basics of modern large scale systems:

Massively Parallel Databases and MapReduce Systems. By Shivnath Babu and Herodotos Herodotou. Foundations and Trends in Databases, 2013

Test 0: We provide a Test 0 that is designed to 1) help you get an idea about how fit you are for the class and 2) bootstrap your C coding skills. Essentially Test 0 consists of an independent data structure design and implementation in C that will allow you to practice basic system design, coding and debugging skills. In addition, several fundamental section videos are posted on the class website about system coding and profiling to help you with that.

GRADING

- Class discussions: 30%
- Paper presentation: 15%
- Semester project: 45%
- Midway check-in: 10%



SEMESTER PROJECT

Each student completes a semester project. There are two kinds of semester projects:

- 1) a systems project, and
- 2) a research project.

Systems projects are tailored to provide background on state-of-the-art systems, data structures, and algorithms. They include a design component and an implementation component in C or C++, addressing low-level system issues such as memory management, hardware-aware processing, parallel processing, read/write trade-offs, and scalability.

The project descriptions can be found on the class website. Projects include building a full NoSQL system and building a full LLM.

This is a focused project that, while it is not extremely heavy in terms of the amount of code you have to write, will bring you up against basic modern system design issues and trade-offs. We will upload a detailed description on the class website before the beginning of the semester. Systems projects will be completed individually; each student will work on their own project.



The research project, on the other hand, is much more tailored to design and proof-of-concept implementations, aiming to solve open problems. Research projects are tailored to give students a taste of research and to lead to publications. When working on a research project, students will work closely with the instructor and members of DASlab on the lab's active research projects. Students will work in groups of three. Such projects are mainly about thinking, reading, and writing, and much less about coding, although proof-of-concept implementations will be our end goal in some cases. Project descriptions are announced every year within the first 3 weeks of class but you can always check the latest publications of DASlab to see the most recent active research topics.

Research projects will be offered to students who have taken CS165 in the past and to those with a significant systems background (e.g., systems PhDs). This will be done in consultation with the instructor.

Systems project can evolve into research projects throughout the semester for students who progress fast and want to continue with research.

What is a successful project? For systems projects, we will specify the functionality and performance metrics you must achieve as part of the project description. For research projects, we will provide specific questions you need to answer when we set up each project.



Evaluation: There are no finals or midterms. At the end of the semester, each student will have a meeting with the instructor and another with the TFs, during which they will demonstrate their projects and answer design questions about them. [Tip: Past experience shows that frequent participation in office hours, brainstorming sessions, and sections means that the instructor and the TFs are very well aware of your system and your progress, which makes the final evaluation a mere formality for these cases.]

Collaboration policy: The systems project is an individual project. The final deliverable should be personal; you must write all the code for your system from scratch, and all documentation and reports. Discussing the design and implementation problems with other students is allowed and encouraged! We will do so in class, during office hours, and during brainstorming sessions. Research projects will be in groups of three. As with the systems project, we encourage cross-team discussions, but ultimately each team should deliver a project that is clearly its own.

Late days policy: All projects are due at the end of the semester, and this is when they will be graded. The more input you give us throughout the semester, the more we can help you learn. In the systems project description, you can find a detailed time schedule that we propose you follow. Similarly, we will set up specific timelines for each research project. All timelines are ideal plans, and you can adjust them to your schedule.



Note: Experience shows that each year, some students cannot handle the freedom to self-pace and end up significantly deviating from the schedule. We will send you frequent reminders, but please note that deviating from the schedule by more than a couple of weeks will most likely mean you will not be able to finish the whole project by the end of the semester (unless you are an experienced systems student).

Midway Check-in: The goal here is to demonstrate that you are having decent progress and mainly to avoid falling behind. By early March, each student working on a systems project should deliver 1) a design document and 2) one performance experiment that demonstrates an early result (10%). A template of the expected design document will be provided early in the semester.

FEEDBACK

We welcome feedback and ideas about the course at any point during the semester. Just come and chat with us during office hours! Tell us how you are keeping up and how we can make it easier for you.



NO LAPTOP/PHONE POLICY

CS265 is based on interaction. We want students actively participating in class and interactive sessions, asking and answering questions to maximize learning. In each class, we will bring a printed copy of the slides for each one of the students so you can follow along and to keep notes on paper. So you do not need your laptop or phones for notes or looking up the slides online.

In fact, recent studies show that even if you only use a laptop for note taking, it can have a negative impact on how well you understand the material in class¹.



[The Pen Is Mightier Than the Keyboard: Advantages of Longhand Over Laptop Note Taking. Pam A. Mueller and Daniel M. Oppenheimer. Psychological Science. 2014, Vol. 25(6) 1159–1168]

¹ There are cases where having a phone or laptop during class is necessary such as when you expect an important call or message or when you need the laptop to better follow the slides due to any issues with your eyes or ears. Just let the instructor know and all such cases will be granted permission to use any tools necessary.

GUEST LECTURES

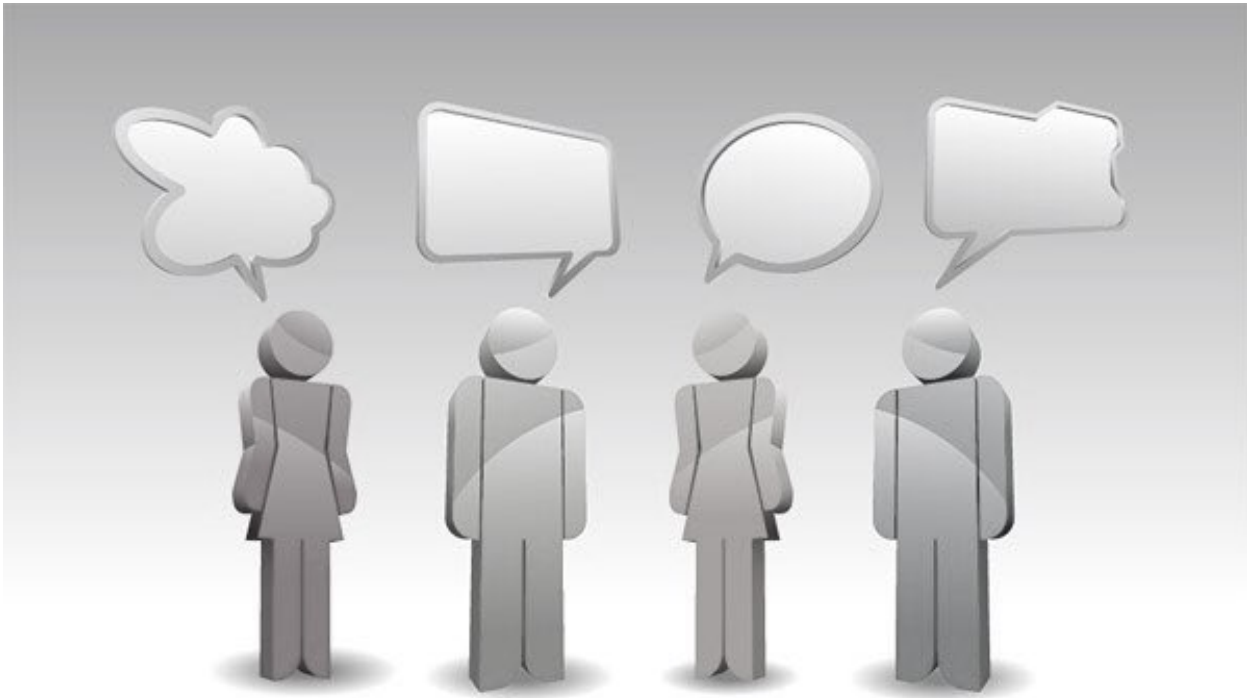
Every semester we arrange a few guest lectures by leaders in data system design from industry and academia. Past guest lecturers in our classes include: Guy Lohman from IBM Research, Erietta Liarou from EPFL Lausanne, Alkis Simitsis and Georgia Koutrika from HP Labs, Nikita Shamgunov from MemSQL, Laura Haas from IBM Research, Nga Tran from Vertica, Jignesh Patel from University of Wisconsin, Johannes Gherke, from Microsoft, Marcin Zukowski from Snowflake, Richard Hipp from SQLite, Ryan Johnson from Logicblox.

You will get the opportunity to both attend a guest lecture and to actively participate in discussions with our guest speakers.



REQUIRED TEXTBOOK

The class is about state-of-the-art data system design. There is no textbook for that. Thus, we use recent research papers and surveys, which will be posted on the course website, which you will have access to through the Harvard network.



ONLINE DISCUSSIONS

We will use ED Discussions as a forum for online discussions. The links are posted on the class website. You are welcome to post any questions that might help you better understand the material or with the project. Anonymous posting (to other students) will be enabled so that students feel more comfortable posting.

BASIC RULES: We have only a few basic rules to keep the forum functional and useful for students while remaining manageable for staff.

- 1) We ask that you search the forum thoroughly before posting a question so we do not have duplicate entries.
- 2) Please make sure to stay on top of all staff posts (especially those that are pinned). Anything we post on the forum is considered “known”.
- 3) Do not use the forum to post code or ask for help with debugging. While it can work in some cases, debugging through messaging is a pain and time-consuming. We have labs every day. Bring your laptop, and we will help you on-site, or join remotely, and we will help you via a shared screen mode.
- 4) Do not use the forum for anything other than technical questions or questions about class logistics. If you want to discuss any concerns about your progress, fit for the class, or anything else you should come to OH.

PLAGIARISM

You are responsible for understanding Harvard and Harvard Extension School policies on academic integrity (www.extension.harvard.edu/resources-policies/student-conduct/academic-integrity) and how to use sources responsibly. Not knowing the rules, misunderstanding the rules, running out of time, submitting "the wrong draft", or being overwhelmed with multiple demands are not acceptable excuses. There are no excuses for failure to uphold academic integrity. To support your learning about academic citation rules, please visit the Harvard Extension School Tips to Avoid Plagiarism (www.extension.harvard.edu/resources-policies/resources/tips-avoid-plagiarism), where you'll find links to the Harvard Guide to Using Sources and two, free, online 15-minute tutorials to test your knowledge of academic citation policy. The tutorials are anonymous open-learning tools.

ACCESSIBILITY

Harvard and the Extension School are committed to providing an accessible academic community. The Disability Services Office offers a variety of accommodations and services to students with documented disabilities. Please visit www.extension.harvard.edu/resources-policies/resources/disability-services-accessibility for more information and do not hesitate to contact prof. Idreos directly, by email, with any questions or concerns you might have.

EXTENSION SCHOOL

This section supplements the basic syllabus with details specific to extension school students.

CS265 is a highly research-oriented course structured very differently from other classes, valuing and promoting critical thinking. For most students, this requires a transition phase. Please check the syllabus and requirements carefully before committing to this course.

In addition, keep in mind that successfully completing this course will require participation in OH and Lab sessions. Even if they are not mandatory, they are critical for students to understand how to think about the material and how to design solutions. Especially if you do not have all the background described in the syllabus (i.e., if you have not taken a research-oriented systems course with a heavy systems project), you should budget time for frequent participation in both Labs and OH, as well as many hours of additional work each week, to build the foundations needed.

Lecture: Lectures will be broadcast live. Lectures will also be available on demand within 24 hours after each class. Students can watch the live or recorded broadcast in their browser. The link to the CS265 broadcasts will be available on the Canvas website and will also be posted on the class website before the first lecture.

Participation: Extension school students will be able to participate live in classes, office hours, and labs via web-conference tools (we will use Zoom). The course staff will be online with Zoom during each session that is marked as “remote” and you will be able to actively interact with the staff. In addition to standard chat and call features, Zoom offers screen sharing, which you can use when you need help with specific issues, such as debugging.

Capturing Discussions: Given that a big portion of the class is based on interaction, extension school in cooperation with the class staff is always working to set-up a system with several microphones across the classroom so we can accurately and clearly capture brainstorming discussions and comments during class time. Microphones will “follow” the instructor.

Grading: Although we encourage extension school students to take advantage of the opportunity to interact with staff and participate in live classes, we know that, for practical reasons, this will not be possible for all remote students. For this reason, extension school students will not receive a “class participation” grade. The rest of the course is exactly the same as what local students do.

For this reason, the portion of the class participation grade (30%) will be distributed to other class components, and the final grade breakdown is as follows: project (60%),

presentation (25%). The 15% for the midway check-in completes the grade distribution for the extension school.

Discussion Forum: Given that remote students often have different needs due to distance, there is a separate forum tailored to extension schools. Look at the class website for the forum link for extension school students.

Office Hours and Labs: Extension school OH and Labs are held remotely on weekends, which is typically more convenient for most students. In this way, we can have more flexibility to accommodate students with day jobs who cannot attend during the week. The schedule will be posted at the beginning of the semester on the class website and the forum.

Starting Date: Note that extension schools usually list the class start date as one day after the actual start date. In fact, this is when the first video will be available. However, extension school students will still be able to stream the first class live on the first day of classes and participate as usual. Check the class website for the exact schedule if you want to participate live.